Flexible Load Scheduling of Hydrogen Energy Data Centers: An Efficiency-Preferred Non-Convex Uncertain Optimization Approach

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***Abstract*—The hydrogen energy data center (HEDC) represents an emerging clean energy infrastructure. Due to its potential load flexibility, HEDC is regarded as a valuable object for power demand response (DR) programs. This paper designs a DR-based scheduling strategy to optimize HEDC operations, achieving simultaneous improvement in electricity cost reduction and renewable energy (RE) utilization. This optimization strategy needs to address three key non-convex challenges: complex load characteristics of HEDCs, dynamic electricity pricing, and RE uncertainty. Focusing on these challenges, a non-convex set interval optimization (NSIO) approach is proposed to obtain high-performance and precise solutions under uncertain conditions. The advantages of the proposed NSIO are that: (*i*) modeling properties of efficiency preference; (*ii*) exact convex relaxation techniques for reducing computational complexity; (*iii*) fully explicit mathematical modeling for achieving global optimality. Comprehensive case studies are conducted on the modified IEEE 24-bus power system. The results validate that the above response strategy effectively improves operational costs and RE utilization. Through systematic evaluation of five state-of-the-art uncertain optimization approaches, the proposed NSIO demonstrates 2.3% $\sim$ 20.0% superior performance compared to classical approaches.**

***Index Terms*—Hydrogen energy, data center (HEDC), demand response (DR), economic scheduling, non-convex uncertain set, interval optimization.**

NOMENCLATURE

Abbreviations

HES	Hydrogen energy system.
HEDC	Hydrogen energy data center.
WP	Wind power.
RE	Renewable energy.
DR	Demand response.
SO	Stochastic optimization.
RO	Robust optimization.
IO	Interval optimization.
NSIO	Non-convex set interval optimization.

Sets and Indices

i	Index of data centers.
j	Index of power system buses.
k	Index of thermal power generators.
w	Index of wind power farms.
t	Index of time slots.

Parameters

$\mathrm{N}_{i,t}^{\mathrm{sv}}$	Operating number of servers.
$P_{\mathrm{st}}^{\mathrm{sv}}$	Static power of server (kW)
k	Power coefficient (kW/GHz^3)
κ	Proportional coefficient (GHz^{-1})
D	Maximum tolerable processing time (s)
PUE	Power utilization efficiency.
$\mathrm{IT}_{\mathrm{i,t}}^{\mathrm{MT}}$, $\mathrm{IT}_{\mathrm{i,t}}^{\mathrm{RT}}$	Migratable and real-time workloads.
N,M,H,W	Number of data centers, power system buses, thermal generators, and wind power farms.
Ω	Supremum constant.
η^{el}, η^{f}	Energy conversion efficiencies of H_2 electrolysis and fuel cells.
η^{ch}, η^{dis}	H_2 transport efficiencies.
ρ	H_2 energy density (Nm ^{3}/(kW$\cdot$ h))
γ^{Ch}, γ^{Disch}	Charge/discharge cost coefficient of HES ($/kW $\cdot$ h)
$\mathrm{D}_{j,t}^{\mathrm{Tr}}$	Conventional load (kW)

Variables

$P_{\mathrm{i,t}}^{\mathrm{sv}}$	Dynamic power of server (kW)
$f_{i,t}$	Average working frequency of server (GHz)
$U_{i,t}$	Utilization rate of server (%)
$\mu_{i,t}^{\mathrm{sv}}$	Available workload quantity.
$\lambda_{i,t}$	Processing data workload quantity.
$\lambda_{i,t}^{\mathrm{MI}}$	Migrating-in workload quantity.

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$P_{i,t}^{\mathrm{CE}}$	Cooling load of data centers (kW)
$P_{i,t}^{\mathrm{Ch}}, P_{i,t}^{\mathrm{Dis}}$	Charging/discharging power of HES (kW)
$Q_{i,t}^{\mathrm{SH_2}}$	Existing H_2 storage (Nm^3)
$\omega_{i,t}^{\mathrm{Ch}}, \omega_{i,t}^{\mathrm{Disch}}$	Intermediate integer variables.
$P_{k,t}^{\mathrm{Th}}, P_{w,t}^{\mathrm{WP}}$	Connected thermal and wind power (kW)
$P_{w,t}^{\mathrm{AWP}}$	Upper bound of available power (kW)
α_t	Real-time electricity prices ($/kW · h)
β_t	Quantitative economic incentives ($)
x	Subjective decision variable.
y	Uncertaint variable.
λ	Dual variable of inequality constraint.
υ	Dual variable of equality constraint.
ξ, δ	Intermediate variables.

I. Introduction

A. Backgrounds

DATA centers are one of the critical infrastructures for smart cities, providing essential computing and data services. These enable intelligent operations across commercial, industrial, and transportation sectors [1], [2]. In China's rapidly expanding digital economy, operational data centers contained over 7.6 million standard racks by 2023, delivering 197 EFLOPS of computing power and 1080 EB of storage capacity. This growth has elevated data centers to major electricity consumers, currently representing over 3% of national consumption with a sustained 30% annual growth rate since 2018 [3]. As a result, data centers have become significant customers on the electricity demand side. Moreover, the automatic management of data center operations creates unique load flexibility, positioning them as valuable assets for demand-side load scheduling. Their ability to dynamically adjust power consumption makes data centers ideal participants in power demand response (DR) programs [4].

Existing research has extensively investigated data center energy management in DR program. Specifically, the management objects includes the power loads of computing server, cooling equipment, and associated energy systems [5]. Notably, Ref. [6] comprehensively quantifies the impact of power demand migration among data centers on carbon reduction in the U.S. power system. For independent power providers PJM and CAISO, this demand migration contributes to reduce the power gap caused by the massive data load of the former, and utilize the excessive renewble energy (RE) from the latter. Ref. [7] proposes a dynamic voltage frequency scaling (DVFS) model of computing server. This fine mathematical model is able to exactly quantify the adjustment ability of data load. In the DR program of multiple data centers, the data load is applied as a flexible resouce on the optimization of power grid transmission losses.

Data centers are not merely load entities, their associated energy systems generally involve RE power stations, thermal power generators, energy storage systems [8], among others. In recent years, hydrogen energy systems (HES) have emerged as a novel solution for the energy storage and supply of data centers. This configuration leverages fuel cell technology to achieve zero-carbon operations and obviously improve the power usage effectiveness of data center. On the other hand, the hydrogen energy system demonstrates notable advantages over conventional lithium battery storage: (*i*) high energy density of hydrogen supports to the sustained and stable power supply; (*ii*) modular design of HES facilitates rapid deployment [9].

In practice, Microsoft data centers have completely replaced diesel generators with hydrogen energy and other energy sources in 2023 [10]. In Chinese metropolitan area planning, intra-regional data centers are encouraged to incorporate hydrogen energy stations as backup systems. [11]. The following are the studies focused on hydrogen energy data centers (HEDCs). Considering various scenarios, Ref. [12] proposes planning and configuration schemes for data centers coupling HES. These schemes focus on achieving reliable bidirectional electric-hydrogen conversion and sufficient power supply within the data centers. From the perspective of electricity control, Ref. [13] presents specific system modeling and corresponding control strategies for HEDCs. In Ref. [14], compared with simple data center loads, multiple HEDCs are regarded as more flexible resources in DR programs, and their effective dispatch significantly enhances the RE utilization of regional power systems. The above practices and research preliminarily validate the feasibility of HEDCs and indicate that HEDC is novel and valuable research object in DR program.

From the perspective of data center operators, the primary objective of its proactive load adjustment is to improve electricity cost efficiency. Therefore, it is necessary to focus on the economic feedback and cost analysis in DR program [2]. Based on a win-win condition for the electricity sector and data center operators, Ref. [15] proposes an economic incentive-based DR program to optimize the load rate of regional substations. In emergency DR programs, Ref. [16] establishes a cooperative game model to allocate optimal cooperative profits for reduced load data centers. Ref. [17] develops a DR-based planning model to configure the most cost-effective energy systems within data centers, involving cooling equipment, heat recovery units, and battery storage systems.

On the other hand, the impact of uncertain factors on the economic efficiency of data centers is significant. In the economic operation optimization, Ref. [18] employs large-scale sampling and scenario reduction techniques to generate representative renewable energy scenarios, which characterize the uncertainties associated with wind and photovoltaic power supply. Ref. [19] proposes a three-stage distributed robust optimization. This approach is applied to address the planning of data centers integrated electricity-heat systems under wind power uncertainty. Focusing on energy consumption optimization for low-carbon data centers, Ref. [20] employs generative adversarial networks to establish RE uncertain sets and designs a two-stage robust optimization. However, these studies rarely address the coupling between hydrogen energy and data centers. For emerging load entities, quantitative analysis of the economic implications and uncertainty impacts in HEDCs is crucial. Furthermore, existing uncertain optimization approach exhibit insufficient decision-making performance in efficiency-preferred problems. These issues will be comprehensively discussed in the next subsection.

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B. Related Works of Uncertain Optimization

Focusing on the field of uncertain optimization, existing methods can be categorized into stochastic optimization (SO) and robust optimization (RO). For classical SO, an accurate probability distribution of uncertain parameters is a prerequisite for its application. High-confidence stochastic scenarios support the implementation of SO, and the sampling and generation of these scenarios need to be based on this probability distribution. For uncertainties involving multiple periods and scattered areas such as RE generation, accurate distribution modeling is challenging to implement due to the excessive requirements of data sets and computational load [21]. For RO, its advantage lies in providing a robust optimal solution that is feasible across all uncertainty scenarios, meaning this solution is constraint-violation-free for any uncertain parameter. Correspondingly, the disadvantage is that the performance of the robust optimal solution is generally conservative [22]. This conservatism arises because the solution must adhere to the strictest constraints. Many related works aim to refine the uncertainty set to improve RO conservatism [23]. Nonetheless, this set refinement increases the possibility of constraint violations, and the basic model nature of RO is for emphasizing feasibility rather than decision performance.

Interval optimization (IO), an efficiency-preferred uncertainty optimization approach, focuses on obtaining the most optimal and stable solutions within uncertain environments. Similar to RO, uncertain model parameters in IO are set as special variables, which are constrained or valued by specific submodels. The feasible region of RO uncertain variables is limited within the most restricted constraints. In contrast, the optimal solution of IO can be based on the complete feasible region of uncertain variables. Essentially, the larger feasible region and the efficiency-preferred optimization objective are the reasons that the expected performance of IO optimal solutions is superior to that of RO solutions. Compared to SO, the established boundaries of uncertainty parameters are easier to obtain than precise probability distributions. This illustrates that IO can relieve the required computational load and practical complexity. The following is an overview of the development of interval optimization.

The initial concept of interval optimization is first introduced in Ref. [24]. Within the uncertain variable interval, the research establishes a bi-objective IO model, using the median and width of the original optimization objective to represent the optimality and stability of decision performance. Notably, the indicators of median and width are formulated based on the optimal and worst-case boundaries of objective performance for a certain decision, i.e., optimistic and pessimistic boundaries. The generation of these two boundaries forms two types of sub-optimization within the main optimization. However, much of the content in the study focuses on conceptual models, rather than providing explicit, solvable mathematical formulations used for modeling and optimization strategies.

For a linear convex economic scheduling, Ref. [25] proposes an absolute value constraint transformation to explicitly formulate the sub-optimization for generating the pessimistic boundary. To further linearize the absolute value constraints, equivalent relaxations are introduced to adapt the application requirements of the branch-and-bound algorithm. In fact, the optimal solution of convex optimization exists at the extreme points within the feasible convex region. For interval variables, the extreme points are the interval boundary values [26]. According to this property, Ref. [27] develops the calculation rules for the optimistic-pessimistic boundaries. These rules simplify the implementation of sub-optimization within the main optimization, thereby reducing computational complexity. However, this convexity-based property and calculation rules are generally not valid for the non-convex optimal solution. Thus, designing solving approaches for non-convex IO is more challenging, especially for the modeling of sub-optimization.

Concentrating on the related issues of non-convex IO, Ref. [28] designs a solving procedure based on a bi-level group search algorithm. In this context, the sub-optimizations are implemented by the algorithm at the lower level. In Ref. [29], an improved multi-objective deep reinforcement learning approach is applied to address non-convex IO, with the sub-optimization modeled based on a Markov decision process. Obviously, these non-mathematical algorithms result in heavier computational loads due to the higher complexity required for modeling non-convex IO. Additionally, these approaches cannot guarantee the global optimality of the convergent solution. In Ref. [30], the solving procedure for non-convex IO is established as a two-stage strategy. The first stage aims to obtain optimistic-pessimistic boundaries through generational error correction, which are then used in the final solution optimization of the second stage. Combining conservative constraints from RO, the fixed-integer decomposition-substitution algorithm is proposed to generate the pessimistic boundary, independent of the main optimization process [31]. However, it is necessary to note that verifying the compatibility between solutions and the mentioned objective boundaries, as well as the global optimality, is challenging in these stepwise non-convex IO approaches. Ref. [14] models the constrained sub-optimizations for generating optimistic-pessimistic boundaries and further proposes a completely mathematical non-convex IO. This corresponds to a complete and explicit model, which is valuable for exploring global optimality. However, the application of this approach is limited to linear sets of uncertainties rather than non-convex sets.

According to the above references, the research gaps are summarized as follows. These are based on the perspectives of research objects and uncertainty optimization approaches, respectively.

1) Configuring hydrogen energy system is a novel and practical solution for the energy storage and supply of data centers. Therefore, it is necessary to conduct research on effective load scheduling. Especially in complex uncertain conditions, the study of effective load scheduling and demand response is not only valuable for economic operation of data centers, but also beneficial for the renewable energy utilization in power systems.
2) An interval optimization approach compatible with non-convex uncertainty is a valuable topic. For common non-mathematical approaches, the main disadvantages are

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excessive algorithm complexity and lack of solution optimality. Existing mathematical approaches lack universal explicit modeling and solution strategies, failing to achieve the generalizability, especially not compatible with non-convex uncertainty models.

C. Contributions

In this paper, the integrated system combining data centers and auxiliary hydrogen energy systems is considered as an adjustable load resource on the power demand side. In the demand response program for renewable energy utilization, quantified uncertain impacts are incorporated into the economic scheduling of hydrogen storage data centers. The corresponding interval optimization is formulated, with both the original model and uncertain sets being non-convex. To address this, equivalent transformations are introduced to linearize the specific non-convex sets. Based on this, modeling and solving approaches for non-convex interval optimization are proposed to obtain the optimal decision set for economic scheduling.

1) The uncertain economic scheduling strategy for hydrogen energy data center (HEDC-UES) is designed to obtain precise and efficient decisions. In the demand response program for renewable energy utilization, this strategy comprehensively quantifies the multi-factor impacts on the energy cost of data centers. These factors includes load adjustment, hydrogen energy application, fluctuations in real-time electricity prices, and uncertain renewable energy supply.
2) The non-convex sets interval optimization (NSIO) approach is developed to solve for the optimal and most stable solutions for the HEDC-UES. This approach corresponds to a complete and mathematically explicit NSIO model, which results in lower computational complexity and ensures the global optimality of the solution set.
3) The linearizations are proposed for the non-convex uncertain sets in the HEDC-UES model. Based on specific optimization, the integer and preference sets are transformed into equivalent convex constraints, forming the basis for the mathematically explicit modeling of NSIO.

The rest of this paper is organized as follows. Section II illustrates the strategy purpose and mathematical modeling of HEDC-UES. Section III introduces the basic IO concept and the specific linearization for HEDC-UES model. On this basis, Section IV formulates the fully explicit mathematical NSIO model and clarify the related theory. Section V aims at the case setup, experiment, and analysis. Eventually, Section VI presents the study conclusions.

II. Economic Scheduling of Hydrogen Energy Data Centers Considering Uncertainties

A. HEDC-UES Strategy Overview

From the power system dispatcher's perspective, efficient renewable energy (RE) utilization represents a critical operational challenge. Inadequate control of RE integration may result in substantial energy curtailment, power quality degradation, and potential system failures [32]. As the RE penetration rate increases, the operational challenges stemming from its inherent uncertainty become increasingly significant. As discussed in Section I, HEDCs with load flexibility serve as both valuable and feasible dispatch resources. Within DR programs, HEDCs perform potentiality in utilizing surplus RE generation and maintaining power balance.

From the data center operator's perspective, the primary objective of managing power loads and energy systems is to minimize energy consumption costs. Recognizing HEDCs as valuable demand-side resources, this paper formulates the cost optimization problem within a demand response (DR) framework. Using WP as a representative case, the DR program offers economic incentives for HEDC operators to accommodate surplus WP generation. As illustrated in Fig. 1, the total energy costs comprise three key components: (1) electricity costs for data center loads, (2) electricity and operational costs for maintaining the HES, and (3) economic benefits from WP utilization incentives.

As shown in the figure, the real-time power load of a data center is determined by the data workload processing volume, server operational status, and cooling equipment performance. The HES exhibits dual characteristics, functioning as both a load and a power source. Specifically, the HES can consume surplus electricity to produce hydrogen, thereby increasing its energy storage, or convert hydrogen back to electricity, reducing the data center's reliance on external power sources. Consequently, the operator can efficiently schedule these flexible components to optimize energy costs. For instance, during periods of low electricity prices or excess WP supply, the operator can increase the HEDC's total power demand to achieve low prices and utilize surplus electricity. Conversely, during periods of high electricity prices and WP shortages, the operator can reduce the total power demand to minimize the use of expensive external electricity. This serves as a simplified example; a detailed analysis of multiple cost factors is required to formulate precise and quantifiable operational strategies.

On the other hand, WP supply exhibits significant uncertainty. In multi-period and multi-scenario dispatch frameworks, quantifying and analyzing the economic incentives for WP utilization poses a major challenge in cost optimization. Furthermore, real-time electricity prices are subject to multiple stochastic factors, including supply-demand dynamics, seasonal and weather variations, renewable energy intermittency, and fuel price volatility [33]. For typical electricity consumers, these price fluctuations are difficult to predict quantitatively, representing an additional source of uncertainty. To address these challenges, this paper introduces a dedicated quantification model to characterize such uncertainties. Building on this foundation, Section IV develops the NSIO approach to derive optimal and robust operational decisions.

Accordingly, the next subsections will first model the loads for the data center, the associated HES, and the regional power system. Additionally, the uncertain models of electricity prices and wind power incentives will be formulated. Finally, a multi-factor uncertainty economic dispatch model for HEDCs will be constructed.

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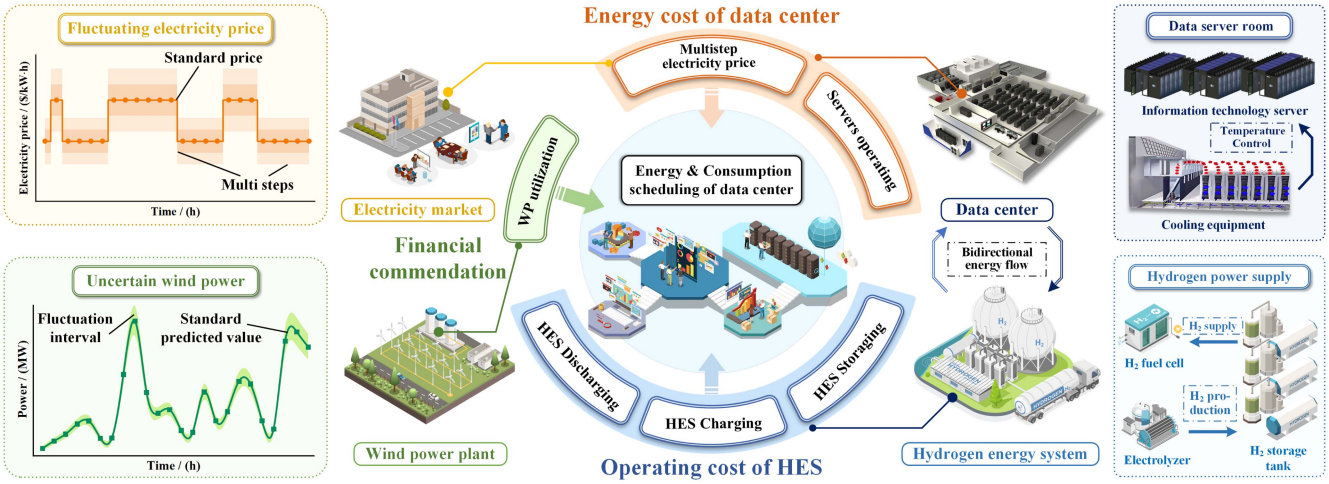


Fig. 1. Illustration for HEDC-UES Strategy.

B. Modeling for Hydrogen Energy Data Center

i) Model of data center server load: The primary source of power consumption in data centers is the server. Server power consumption can be categorized into: (*i*) static power $P_{\mathrm{st}}^{\mathrm{sv}}$, which maintains the regular operation of the unit; (*ii*) dynamic power, which is utilized for performing computational workloads, and is adjustable load as presented in (1). Specifically, the dynamic power depends on the power coefficient k, the average working frequency $f_{i,t}$ of the servers, and the utilization rate $U_{i,t}$. Specifically, $U_{i,t}$ is formulated based on the processing workload quantity $\lambda_{i,t}$ and the available workload quantity $\mu_{i,t}^{\mathrm{sv}}$.

$$P_{i,t}^{\mathrm{sv}} = \mathrm{N}_{i,t}^{\mathrm{sv}} \cdot P_{\mathrm{st}}^{\mathrm{sv}} + \mathrm{k} \cdot f_{i,t}^{3} \cdot U_{i,t},\ U_{i,t} = \lambda_{i,t}/\mu_{i,t}^{\mathrm{sv}}, \quad (1)$$

where the subscripts $[\cdot]_i$ and $[\cdot]_t$ correspond to the ith data center and t^{th} time period, respectively; $\mathrm{N}_{i,t}^{\mathrm{sv}}$ represents the operating number of servers. Higher working frequency indicates stronger computing power; Thus, the available workload quantity $\mu_{i,t}^{\mathrm{sv}}$ is proportional to the working frequency $f_{i,t}$. This mathematical relationship and the boundary constraint of working frequency are shown in (2).

$$\mu_{i,t}^{\mathrm{sv}} = \kappa f_{i,t}\ ,\quad f_{i,t}^{\min} \le f_{i,t} \le f_{i,t}^{\max}, \quad (2)$$

where κ is the proportional coefficient. The frequencies $f_{i,t}^{\min}$ and $f_{i,t}^{\max}$ denote the lower and upper bounds, respectively.

Among multiple data centers, data workloads are able to implement limited real-time transmission. This is also recognized as a load regulation strategy and should be incorporated into quantitative models for processing data workload $\lambda_{i,t}$. Regarding the limitations of workload migration, the migrating-out workload $-\lambda_{i,t}^{\mathrm{MI}}$ should be less than the migratable workload $\mathrm{IT}_{\mathrm{i,t}}^{\mathrm{MT}}$, and the migrating-in and migrating-out workloads should be balanced. The specific constraints are presented in (3).

$$\lambda_{i,t} = \lambda_{i,t}^{\mathrm{MI}} + \mathrm{IT}_{\mathrm{i,t}}^{\mathrm{MT}} + \mathrm{IT}_{\mathrm{i,t}}^{\mathrm{RT}},\ \lambda_{\mathrm{i,t}}^{\mathrm{MI}} + \mathrm{IT}_{\mathrm{i,t}}^{\mathrm{MT}} \ge 0,\ \sum_{\mathrm{i=1}}^{\mathrm{N}} \lambda_{\mathrm{i,t}}^{\mathrm{MI}} = 0, \quad (3)$$

where $\lambda_{i,t}^{\mathrm{MI}}$ denotes the migrating-in workload; $\mathrm{IT}_{\mathrm{i,t}}^{\mathrm{MT}}$ and $\mathrm{IT}_{\mathrm{i,t}}^{\mathrm{RT}}$ represent the migratable and real-time workloads, respectively; N is the number of data centers. The total time for queuing $T_{i,t}^{\mathrm{Q}}$ and processing $T_{i,t}^{\mathrm{P}}$ not exceed the specified threshold, otherwise it will degrade the service quality of data center. This is a mandatory constraint, thus $T_{i,t}^{\mathrm{Q}}$ and $T_{i,t}^{\mathrm{P}}$ are limited as (4).

$$T_{i,t}^{\mathrm{Q}} + T_{i,t}^{\mathrm{P,max}} \le \mathrm{D}\ ,\ T_{i,t}^{\mathrm{Q}} = \frac{1}{\mu_{i,t}^{\mathrm{sv}} \cdot \mathrm{N}_{i,t}^{\mathrm{sv}} - \lambda_{i,t}},\ T_{i,t}^{\mathrm{P}} = \frac{1}{\mu_{i,t}^{\mathrm{sv}}}\ , \quad (4)$$

where D denotes the maximum total time and $T_{i,t}^{\mathrm{P,max}} = (\kappa f_{i,t}^{\min})^{-1}$.

ii) Model of cooling equipment load: The operation of data center servers generates a significant amount of heat. Next-generation data centers are actively adopting liquid cooling systems to regulate server temperatures, a configuration that significantly reduces refrigeration costs and decreases overall energy consumption. Thus, in addition to the server load, the cooling load $P_{i,t}^{\mathrm{CE}}$ represents the maximum power demand for a data center. A model based on power utilization efficiency (PUE) is used to establish the cooling load model [18]. Consequently, the total load $P_{i,t}^{\mathrm{DC}}$ is formulated as shown in (5).

$$P_{i,t}^{\mathrm{CE}} = (\mathrm{PUE} - 1) \cdot P_{i,t}^{\mathrm{sv}},\ P_{i,t}^{\mathrm{DC}} = \mathrm{PUE} \cdot P_{i,t}^{\mathrm{sv}}. \quad (5)$$

iii) Model of hydrogen energy ststem: The HES exhibits both power load and source characteristics, enabling bidirectional energy conversion between electricity and hydrogen ($\mathrm{H_2}$). Accordingly, the formulations for the charging power $P_{i,t}^{\mathrm{Ch}}$, the discharging power $P_{i,t}^{\mathrm{Dis}}$, and the existing $\mathrm{H_2}$ storage $Q_{i,t}^{\mathrm{SH_2}}$ are presented in (6) and (7).

$$P_{i,t}^{\mathrm{Ch}} = Q_{i,t}^{\mathrm{H_2}(+)}/(\eta^{\mathrm{el}} \cdot \rho),\ \ P_{i,t}^{\mathrm{Dis}} = Q_{i,t}^{\mathrm{H_2}(-)} \cdot \eta^{\mathrm{f}}/\mu\ , \quad (6)$$

$$Q_{i,t}^{\mathrm{SH_2}} = Q_{i,t-1}^{\mathrm{SH_2}} + (\eta^{\mathrm{ch}} \cdot Q_{i,t}^{\mathrm{H_2}(+)} - Q_{i,t}^{\mathrm{H_2}(-)}/\eta^{\mathrm{dis}}) \cdot \Delta\mathrm{t}, \quad (7)$$

where $Q_{i,t}^{\mathrm{H_2}(+)}$ and $Q_{i,t}^{\mathrm{H_2}(-)}$ denote the production and consumption of $\mathrm{H_2}$, respectively; η^{el} and η^{f} are the energy conversion efficiencies of $\mathrm{H_2}$ electrolysis and fuel cells, respectively; ρ represents the electricity consumption per unit of $\mathrm{H_2}$ generated;

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μ represents the electricity generation per unit of H_2 consumed; η^{ch} and η^{dis} are the H_2 transport efficiencies for the charging and discharging processes, respectively; and Δt denotes the time slot. The boundary constraints for the charging and discharging power are shown in (8).

$$P_{i,t}^{\mathrm{Ch,min}} \le P_{i,t}^{\mathrm{Ch}} \le P_{i,t}^{\mathrm{Ch,max}}, P_{i,t}^{\mathrm{Dis,min}} \le P_{i,t}^{\mathrm{Dis}} \le P_{i,t}^{\mathrm{Dis,max}}. \tag{8}$$

The HES cannot simultaneously function as both a load and a power source; thus, charging and discharging processes cannot occur concurrently. Additionally, due to stringent power quality requirements, the electricity generated by the HES is used exclusively for self-supply to the data center and is not connected to the power system. The specific constraints are shown in (9).

$$P_{i,t}^{\mathrm{Ch}} \le \omega_{i,t}^{\mathrm{Ch}} \cdot \Omega, \quad P_{i,t}^{\mathrm{Dis}} \le \omega_{i,t}^{\mathrm{Disch}} \cdot \Omega, \quad P_{i,t}^{\mathrm{Dis}} \le P_{i,t}^{\mathrm{DC}}, \tag{9}$$

where $\omega_{i,t}^{\mathrm{Ch}}$ and $\omega_{i,t}^{\mathrm{Disch}}$ represent the charge and discharge status, 0-1 variables, and $\omega_{i,t}^{\mathrm{Ch}} = 1 - \omega_{i,t}^{\mathrm{Disch}}$; Ω is a supremum constant.

C. Power System Modeling

In this subsection, both thermal power $P_{k,t}^{\mathrm{Th}}$ and wind power $P_{w,t}^{\mathrm{WP}}$ are regarded as the main supply sources of the power system. In addition to the HEDC load, the traditional load $\mathrm{D}_{j,t}^{\mathrm{Tr}}$ is also considered here. Thus, the power balance constraint is presented as (10).

$$\sum_{i=1}^{\mathrm{N}} P_{i,t}^{\mathrm{HEDC}} + \sum_{j=1}^{\mathrm{M}} \mathrm{D}_{j,t}^{\mathrm{Tr}} = \sum_{k=1}^{\mathrm{H}} P_{k,t}^{\mathrm{Th}} + \sum_{w=1}^{\mathrm{W}} P_{w,t}^{\mathrm{WP}}, \tag{10}$$

where the subscripts $[\cdot]_j$, $[\cdot]_k$, and $[\cdot]_w$ correspond to the jth bus of the power system, the kth thermal generator, and the wth wind power plant, respectively. The constants M, H, and W represent the number of power system buses, thermal generators, and wind power plants, respectively. $P_{i,t}^{\mathrm{HEDC}}$ denotes the power load of the HEDC, where $P_{i,t}^{\mathrm{HEDC}} = P_{i,t}^{\mathrm{DC}} + P_{i,t}^{\mathrm{Ch}} - P_{i,t}^{\mathrm{Dis}}$. In this context, the model of direct-current power flow is formulated to quantify the power transmission characteristics, and the bus power $P_{m,t}^{\mathrm{G}}$ is established as (11).

$$P_{m,t}^{\mathrm{G}} = \mathrm{D}_{j,t}^{\mathrm{Tr}} + \sum_{i=1}^{\mathrm{N}} \mathrm{s}_{i,m}^{\mathrm{DC}} \cdot P_{i,t}^{\mathrm{HEDC}} - \sum_{w=1}^{\mathrm{W}} \mathrm{s}_{w,m}^{\mathrm{WP}} \cdot P_{w,t}^{\mathrm{WP}} + \sum_{n=1}^{\mathrm{M}} P_{n,t}^{\mathrm{B}}, \tag{11}$$

where $s_{i,m}^{\mathrm{DC}}$ and $s_{i,m}^{\mathrm{WP}}$ are the established binary values defining the bus m connection status of HEDCs and WP plants; $P_{n,t}^{\mathrm{B}}$ represents the inflow power from branches, $P_{n,t}^{\mathrm{B}} = \mathrm{B}_{mn} \cdot (\theta_m - \theta_n)$; B_{mn} denotes the susceptance of branch q, which is between buses m and n; θ_m and θ_n are the voltage phase angles of buses m and n. Furthermore, the constraints involving the thermal power boundary, the thermal power climbing rate, and the transmission security are also considered in this model. For the WP connection, the upper bound of available power $P_{w,t}^{\mathrm{AWP}}$ fluctuates due to environmental factors. In this paper, this uncertain fluctuation is set within an established interval. Thus, the boundaries of power output and the uncertain upper bound of available WP are presented as (12).

$$P_{w,t}^{\mathrm{WP,min}} \le P_{w,t}^{\mathrm{WP}} \le P_{w,t}^{\mathrm{AWP}}, \ P_{w,t}^{\mathrm{AWP}} \in \left[P_{w,t}^{\mathrm{AWP,L}}, P_{w,t}^{\mathrm{AWP,U}}\right], \tag{12}$$

where $P_{w,t}^{\mathrm{WP,min}}$ is the lower power bound of WP output; $[P_{w,t}^{\mathrm{AWP,L}}, P_{w,t}^{\mathrm{AWP,U}}]$ is the uncertain interval of available WP $P_{w,t}^{\mathrm{AWP}}$.

D. Modeling of Uncertain Economic Scheduling

In this subsection, the uncertain models involving the real-time electricity price and the utilization incentive of wind power are first established. On this basis, the optimization model for uncertain economic scheduling is formulated.

i) Model of real-time electricity price: As mentioned in Subsection A, the fluctuations of real-time electricity prices are challenging to predict for HEDC operators. Thus, the values of real-time electricity prices α_t are set within an uncertain set, a discrete interval, as presented in (13).

$$\alpha_t = \{\alpha_t^1, \alpha_t^2, \ldots, \alpha_t^{n-1}, \alpha_t^{n,\max}\}. \tag{13}$$

It is necessary to emphasize that these uncertainties are not completely irregular. They should follow a tiered pricing policy for charging general large electricity customers [15]. Thus, the constraint of this pricing policy is formulated as (14). In this context, higher electricity consumption by HEDCs means more expensive charging standards, i.e., a greater lower bound of uncertainty fluctuations.

$$\alpha_t \ge \alpha_t^1 + (\alpha_t^{n,\max} - \alpha_t^1) \cdot L_{i,t}^{\mathrm{Rate}}, \tag{14}$$

where $L_{i,t}^{\mathrm{Rate}}$ denotes the load rate for HEDC buses, $L_{i,t}^{\mathrm{Rate}} = \sum_{i=1}^{\mathrm{N}} P_{i,t}^{\mathrm{HEDC}} / \sum_{i=1}^{\mathrm{N}} P_{i,t}^{\mathrm{Bus,max}}$; $P_{i,t}^{\mathrm{Bus}}$ is the maximal power supply.

ii) Model of economic incentive for wind power utilization: Quantitative economic incentives β_t should be proportional to the increasing value of wind power utilization, $\Delta P_{w,t}^{\mathrm{WP}}$. On the other hand, the available supply of WP is uncertain, and any shortage leading to utilization reduction should not result in a negative economic incentive. Thus, the value of economic incentives follows non-negative rules, which is specifically formulated as a non-convex uncertain preference set, as presented in (15) [33].

$$\beta_t = \max\{0, \beta \cdot \Delta P_{w,t}^{\mathrm{WP}}\}, \Delta P_{w,t}^{\mathrm{WP}} = \sum_{w=1}^{\mathrm{W}} (P_{w,t}^{\mathrm{WP}} - P_{w,t}^{\mathrm{WP,Pre}}) \tag{15}$$

where β denotes the incentive value for per unit power; $P_{w,t}^{\mathrm{WP,Pre}}$ represents the expected connection of wind power.

iii) Uncertain economic scheduling optimization for HEDC: The optimization objective of the uncertain economic scheduling primarily involves the electricity costs $F_1(x, y_1)$ of the HEDC load, economic incentives $F_2(x, y_2)$ for WP utilization, and the charging and discharging costs and storage losses $F_3(x)$

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of HES. The general optimization model is expressed as:

$$\begin{aligned}
&\min_{x} \quad F(x,y) = F_1(x,y_1) - F_2(x,y_2) + F_3(x)\\
&\text{s.t.} \quad F_1(x,y_1) = \sum_{i=1}^{\mathrm{N}}\sum_{t=1}^{\mathrm{T}} \alpha_t \cdot P_{i,t}^{\mathrm{HEDC}} \cdot \Delta\mathrm{t},\\
&F_2(x,y_2) = \sum_{t=1}^{\mathrm{T}} \beta_t = |\max\{0, x - x_0\}|_1,\\
&F_3(x) = \sum_{i=1}^{\mathrm{N}}\sum_{t=1}^{\mathrm{T}} (\gamma^{\mathrm{Ch}} P_{i,t}^{\mathrm{Ch}} + \gamma^{\mathrm{Dis}} P_{i,t}^{\mathrm{Dis}}) \cdot \Delta\mathrm{t} + \eta^{\mathrm{ch}}\gamma^{\mathrm{S}} Q_{i,t}^{\mathrm{SH_2}},\\
&h(x,y) = 0,\ g(x,y) \le 0,\ y = [y_1, y_2]\\
&y_1 \in \{y_1^1,\ y_1^2,\ \ldots,\ y_1^{n-1},\ y_1^n\},\ y_2 \in [y^{\mathrm{L}}, y^{\mathrm{U}}],
\end{aligned} \tag{16}$$

where γ^{Ch} indicates the cost for generating per unit $\mathrm{H_2}$, and $\gamma^{\mathrm{Ch}} P_{i,t}^{\mathrm{Ch}} \cdot \Delta\mathrm{t}$ is the $\mathrm{H_2}$ production cost; γ^{Dis} denotes the power-supply cost of per unit $\mathrm{H_2}$, and $\gamma^{\mathrm{Dis}} P_{i,t}^{\mathrm{Dis}} \cdot \Delta\mathrm{t}$ is the $\mathrm{H_2}$ power generation cost; γ^{S} represents the cost of storage loss for per unit $\mathrm{H_2}$, and $\eta^{\mathrm{ch}}\gamma^{\mathrm{S}} Q_{i,t}^{\mathrm{SH_2}}$ is the storage loss cost. In the model of (16), $h(x,y)$ and $g(x,y)$ are the equality and inequality constraints, which include the mathematical formulas (1)–(15); x is the decision variables, $x = [f_{i,t}, \lambda_{i,t}^{\mathrm{MI}}, P_{i,t}^{\mathrm{Ch}}, P_{i,t}^{\mathrm{Dis}}, P_{k,t}^{\mathrm{Th}}, P_{w,t}^{\mathrm{WP}}]$; y represents the uncertain variables, $[y_1, y_2] = [\alpha_t, P_{w,t}^{\mathrm{WSP}}]$. y_1 is within the non-convex discrete set, and y_2 is within the interval set with the non-convex preferred constraint. For the discrete set, it is necessary to be equivalently reformulated as the binary set, which is mathematically explicit. Specifically, a group of vectors of binary variable $\{\delta_1^1, \delta_1^2, \ldots, \delta_1^{n-1}, \delta_1^n\}$ is introduced to replace the original uncertain variable y_1. The equivalent transformation is presented as (17).

$$y_1 = \sum \delta_1^n \circ y_1^n,\ \ \sum \delta_1^n = 1,\ \ \delta_1^n \in \{0,1\} \tag{17}$$

where y_1 is equal to the cumulative sum of the Hartman products $\circ$ of all terms y_1^n. With the constraints of $\sum \delta_1^n = 1$ and $\delta_1^1 \in \{0,1\}$, it means that the only one item within the set $\{y_1^1,\ y_1^2,\ \ldots,\ y_1^{n-1},\ y_1^n\}$ are effectively assigned to y_1. For instance, if $\delta_1^1 = 1$, $\delta_1^2 = \delta_1^3 = \ldots = \delta_1^n = 0$, $y_1 = \sum \delta_1^n \circ y_1^n = \delta_1^1 \circ y_1^1$. On this basis, the general formula of $F_1(x, y_1)$ could be further reformulated as (18).

$$\begin{aligned}
\mathrm{Obj}: \quad & F_1(x, \delta_1^1,\ \delta_1^2,\ \ldots,\ \delta_1^{n-1},\ \delta_1^n) = f_1(x)^{\top} \cdot \delta_1\\
&\text{s.t.} \quad \mathrm{A}(x) \cdot \delta_1 \le b, \delta_1 \circ (\delta_1 - 1) = 0,\\
&f_1(x) = [f_1^1,\ f_1^2, \ldots,\ f_1^{\mathrm{N}\times n}]^{\top},\\
&\mathrm{A}(x) = [a_1,\ a_2,\ \ldots,\ a_{\mathrm{N}\times n}],\\
&\delta_1 = [\delta_1^{1\top},\ \delta_1^{2\top},\ \ldots,\ \delta_1^{n-1\top},\ \delta_1^{n\top}]^{\top}
\end{aligned} \tag{18}$$

where $f_1(x)$ and $\mathrm{A}(x)$ represent the objective coefficient and constrained coefficient matrixs, which only include decision x and constants, and b is the constant vector.

III. Interval Optimization and Specific Linearizations

The previous section introduces the HEDC-UES strategy and object modeling. The following sections focus on the modeling and implementation of interval optimization theory applied this strategy.

A. Conceptual Model of Interval Optimization

Interval optimization is an efficiency-preferred decision approach, and its expected solution x is corresponding to the optimal and most stable decision performance considering uncertain factors y_1 and y_2. These ideas are implemented based on optimization for *Median* $M(x, y_1, y_2)$ and *Width* $W(x, y_1, y_2)$ of objective value, as formulated in (19) and (20).

$$M(x,y_1,y_2) = \{\max_{y_1,y_2}[F(x,y_1,y_2)] + \min_{y_1,y_2}[F(x,y_1,y_2)]\}/2 \tag{19}$$

$$W(x,y_1,y_2) = \{\max_{y_1,y_2}[F(x,y_1,y_2)] - \min_{y_1,y_2}[F(x,y_1,y_2)]\}/2 \tag{20}$$

where $\max_{y_1,y_2}[F(x,y_1,y_2)]$ and $\min_{y_1,y_2}[F(x,y_1,y_2)]$ are the sub-optimization problems. For a given decision x, $\min_{y_1,y_2}[F(x,y_1,y_2)]$ is an optmistic value, and denotes the best decision performance within the uncertain intervals of y_1 and y_2. On the other hand, $\max_{y_1,y_2}[F(x,y_1,y_2)]$ is a pessimistic value, and denotes the worst decision performance within the uncertain intervals of y_1 and y_2.

i) The *Median* of the optimistic and pessimistic values indicates the **decision efficiency**. Specifically, in uncertain environments, a solution with a greater objective *Median* value corresponds to better decision performance. Thus, the optimal $M(x, y_1, y_2)$ corresponds to the solution x that yields the best objective expected value under uncertainty.

ii) The *Width* measures the **stability of decision efficiency**. Specifically, in uncertain environments, a solution with a smaller objective *Width* value corresponds to a more stable decision performance. Thus, the optimal $W(x, y_1, y_2)$ corresponds to the solution x that has the minimum fluctuation boundary of the objective value.

In summary, the goal of interval optimization is to obtain a solution that delivers both the best average efficiency and the most stable efficiency. From the practical perspective, mathematically modeling and solving optimization problems with these complex properties is challenging, especially when both the original problem and uncertain factors involve non-convexity.

B. Linearizations for Specific Sub-Optimization

For the uncertain models in (16) and (17), the real-time electricity price is modeled with an equivalent binary uncertain set that follows the tiered policy constraint, while the available wind power involves a preference constraint. These factors introduce non-convexity into the uncertain variables y_1 and y_2. On the other hand, uncertainty modeling can be divided into two types of sub-optimizations. This separation is due to the fact that y_1 and y_2 only separately exist in the objectives $F_1(x, y_1)$ and $F_2(x, y_2)$, with no constraint correlation between each other. In this subsection, the equivalent linearization methods for non-convex sets are designed based on the properties of specific sub-optimization problems. These methods also form the foundation for the explicit IO mathematical modeling.

The complete equivalent linearization derivation is presented as follows: (*i*) An equivalent transformation for Model (18) is

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formulated based on duality principles in **Linearization 1**. In this context, binary variables $(\delta_1^1, \delta_1^2, \ldots, \delta_1^{n-1}, \delta_1^n)$ are replaced by continuous convex variables ${}^1\lambda_1, \xi$; In (*ii*) and (*iii*), leveraging the optimization properties of convex hull extreme points, equivalent linear constraints are constructed to replace non-convex prefered-set constraints. **Linearization 2** and **3** demonstrate the modeling results of this equivalent substitution for both maximization and minimization directions.

i) Linearization for binary uncertain set

Linearization 1: For the maximized or minimized optimization of binary objective $F_1(x, \delta_1^1, \delta_1^2, \ldots, \delta_1^{n-1}, \delta_1^n)|_{x=x_0}$, the vectors of dual variable ${}^1\lambda_1$ and intermediate variable ξ, $\xi = [\xi_1, \xi_2, \ldots, \xi_{\mathrm{N}\times n}]^\top$, are introduced. Accordingly, the linear-convex dual optimization is formulated as (21).

$$\begin{aligned} \max_{{}^1\lambda_1} \quad & F_1(x_0, {}^1\lambda_1, \xi) = -b^\top \cdot {}^1\lambda_1 + \sum_{\xi_i \in \xi} \xi_i \\ \text{s.t.} \quad & \xi_i \le \pm f_1^i + a_i^\top {}^1\lambda_1,\ \xi_i \le 0 \\ & {}^1\lambda_1 \ge 0,\ \ i = 1, 2, \ldots, \mathrm{N} \times n. \end{aligned} \tag{21}$$

Proof: The specific proof is visible in Appendix A. ■

Linearization 1 is the equivalently dual and convex transformation for specific binary optimization, where desicion x is parameterized; The binary variables $\{\delta_1^1, \delta_1^2, \ldots, \delta_1^{n-1}, \delta_1^n\}$ are replaced by continued variables ${}^1\lambda_1$ and ξ in the linear-convex dual model; The optimal original objective value is equal to the optimal dual objective value.

ii) Linearization for preferenced uncertain constraints

Linearization 2: For the minimized optimization of the pessimistic objective $\max_{y_2}[-F_2(x, y_2)]|_{x=x_0}$, the vector of intermediate variable δ_2^1 is introduced. Accordingly, the linear-convex equivalent optimization is formulated as (22).

$$\begin{aligned} \min_{y_2} \quad & F_2(x_0, y_2) = |\delta_2^1|_1 \\ \text{s.t.} \quad & \delta_2^1 \ge 0,\ \ \delta_2^1 \ge x - x_0 \\ & h_2(x, y_2) = 0,\ g_2(x, y_2) \le 0,\ y_2 \in [y^{\mathrm{L}}, y^{\mathrm{U}}] \end{aligned} \tag{22}$$

Proof: The specific proof is visible in Appendix A. ■

Linearization 3: For the maximized optimization of the optimistic objective $\min_{y_2}[-F_2(x, y_2)]|_{x=x_0}$, the vectors of intermediate variable δ_2^2 and δ_2^3 are introduced, and the *Lagrange* function $\mathscr{L}_2$ for variable δ_2^2 is defined as (23).

$$\mathscr{L}_2 = -|\delta_2^2|_1 + {}^1\lambda_3{}^\top \cdot (x - x_0) + {}^2\lambda_3{}^\top \cdot (x_0 - x). \tag{23}$$

where ${}^1\lambda_3$ and ${}^2\lambda_3$ are the introduced dual variables. Accordingly, the linear-convex equivalent optimization is formulated as (24).

$$\begin{aligned} \max_{y_2} \quad & F_2(x_0, y_2) = |\delta_2^3|_1 \\ \text{s.t.} \quad & \nabla_{\delta_2^2} \mathscr{L}_2 = 0, {}^1\lambda_3, {}^2\lambda_3 \ge 0, \\ & {}^1\lambda_3 \circ (\delta_2^2 + x - x_0) = 0, {}^2\lambda_3 \circ (\delta_2^2 + x_0 - x) = 0, \\ & \delta_2^2 \le x - x_0, \delta_2^2 \le x_0 - x, \delta_2^3 \le x - x_0 - \delta_2^2, \delta_2^3 \le -\delta_2^2 \\ & h_2(x, y_2) = 0,\ g_2(x, y_2) \le 0,\ y_2 \in [y^{\mathrm{L}}, y^{\mathrm{U}}] \end{aligned} \tag{24}$$

Proof: The specific proof is visible in Appendix A. ■

Linearization 2 and 3 are the equivalent mathematical model of preferenced constraint $\sum_{t=1}^{\mathrm{T}} \beta_t = |\max\{0, x - x_0\}|_1$, which are also explicit and linear-convex. It is worth noting that the necessary condition for the equivalence to hold is the specific optimization direction. For instance, as minimizing optimization of (22) illustrated in Fig. 2(a), if both the vector elements of $[x - x_0]_1$ and $[x - x_0]_2$ are greater than 0, the feasible region of δ_2^1 is the blue region, i.e., $\delta_2^1 \ge x - x_0$. Thus, the minimum value of the norm $|\delta_2^1|_1$ is valued in the extreme point within the feasible region, $|\delta_2^1|_1 = |x - x_0|_1$, which is equivalent to the value result of the preferenced constraint.

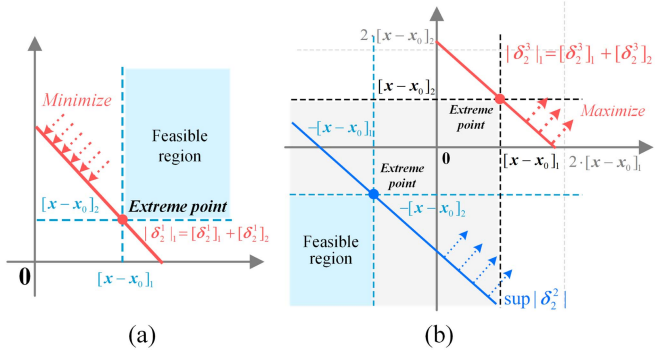


Fig. 2. Illustration of preferenced constraint linearization. (a) The equivalence for minimized optimization; (b) The equivalence for maximized optimization.

As maximizing optimization of (23) illustrated in Fig. 2(b), if both the vector elements of $[x - x_0]_1$ and $[x - x_0]_2$ are greater than 0, the feasible region of δ_2^2 is the blue region, i.e., $\delta_2^2 \le x_0 - x$. Thus, the value of the norm $\sup|\delta_2^2|_1$ is valued in the extreme point within the feasible region, $|\delta_2^2|_1 = |x_0 - x|_1$. On this basis, the constraints for δ_2^3 are equivalent as $\delta_2^3 \le 2 \cdot (x - x_0), \delta_2^3 \le x - x_0$, which means the feasible region of δ_2^3 is the gray area. Thus, the maximum value of the norm $|\delta_2^3|_1$ is valued in the extreme point within the feasible region, $|\delta_2^3|_1 = |x - x_0|_1$, which is equivalent to the value result of the preferenced constraint.

IV. Non-Convex Sets Interval Optimization Approach

In this section, the non-convex set interval optimization (NSIO) approach is proposed. The linearizations discussed in Section III form the foundation for the modeling. Building on this, the concept of constrained sub-optimization enables the fully explicit mathematical formulation of NSIO, resulting in a mixed-integer linear model. The completeness and explicitness of this model ensure both low computational complexity and global optimality of the solution. Consequently, the NSIO approach offers significant advantages, including high computational efficiency and superior solution performance.

Specifically, according to the conceptual IO model, the pessimistic and optimistic sub-optimizations are respectively established based on the two types of uncertain variables, $[y_1^1, y_2^1]$ and $[y_1^2, y_2^2]$. Thus, the median and width could be reformulated as (25)–(26).

$$M(x, y_1^1, y_2^1, y_1^2, y_2^2) = [F(x, y_1^1, y_2^1) + F(x, y_1^2, y_2^2)]/2 \tag{25}$$

$$W(x, y_1^1, y_2^1, y_1^2, y_2^2) = [F(x, y_1^1, y_2^1) - F(x, y_1^2, y_2^2)]/2 \tag{26}$$

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At first, explanation of the constrained transformation is presented based on the general formulas, and the specific model would be visible later. For the sub-optimization, the *Lagrange* function is formulated as (27).

$$\mathscr{L} = \pm F(x,y) + \upsilon^{\top} \cdot h(x,y) + \lambda^{\top} \cdot g(x,y), \tag{27}$$

where υ and λ are the dual variables corresponding to constraints; $\pm$ corresponds to the sub-optimizations of minimization or maximization, respectively. On this basis, *Karush-Kuhn-Tucker* (KKT) condition is introduced only for variable y [26], and the optimality constraints could be formulated as (28)–(29).

$$h(x,y) = 0,\ g(x,y) \le 0,\ \lambda \ge 0, \tag{28}$$

$$\nabla_y \mathscr{L}_y = 0,\ \lambda \circ g(x,y) = 0, \tag{29}$$

The above is the general constraint formulas of constrained sub-optimization. The specific constrained formulas are visible as follows.

Constrained sub-optimization 1: For the maximized or minimized optimization of binary objective $F_1(x, \delta_1^1, \delta_1^2, \ldots, \delta_1^{n-1}, \delta_1^n)|_{x=x_0}$, the *Lagrange* function $\mathscr{L}_3$ is formulated as (30).

$$\begin{aligned}\mathscr{L}_3 = &-b^{\top\,1}\lambda_1 + \sum_{\xi_i \in \xi} \xi_i + \sum_{i \in N \times n} {}^{2}\lambda_1^{i\,\top} \cdot (\xi_i \pm f_1^i - a_i^{\top} \lambda_1^1) \\ &+ {}^{3}\lambda_1{}^{\top} \cdot \xi - {}^{1}\lambda_1{}^{\top} \cdot {}^{4}\lambda_1\end{aligned} \tag{30}$$

Accordingly, the specific optimility constraints of sub-optimization 1 are formulated as (31)–(32).

$$\nabla_{{}^{1}\lambda_1, \xi} \mathscr{L}_3 = 0,\ \xi_i \le f_1^i + a_i^{\top\,1}\lambda_1,\ \xi_i \le 0,\ {}^{1}\lambda_1, {}^{2}\lambda_1, {}^{3}\lambda_1, {}^{4}\lambda_1 \ge 0 \tag{31}$$

$${}^{1}\lambda_1 \circ {}^{4}\lambda_1 = 0,\ {}^{3}\lambda_1 \circ \xi = 0,\ {}^{2}\lambda_1^i \circ (\xi_i \pm f_1^i - a_i^{\top} \lambda_1^1) = 0 \tag{32}$$

where ${}^{2}\lambda_1$, ${}^{3}\lambda_1$, and ${}^{4}\lambda_1$ are the dual variables corresponding to inequality constraints.

Constrained sub-optimization 2: For the maximized optimization of the optimistic objective $\min_{y_2}[-F_2(x,y_2)]|_{x=x_0}$, the *Lagrange* function $\mathscr{L}_4$ is formulated as (33).

$$\begin{aligned}\mathscr{L}_4 = &|\delta_2^1|_1 + \upsilon_2^{\top} \cdot h_2(x,y_2) + {}^{1}\lambda_2{}^{\top} \cdot g_2(x,y_2) - {}^{2}\lambda_2{}^{\top} \cdot \delta_2^1 \\ &+ {}^{3}\lambda_2{}^{\top} \cdot (x - x_0 - \delta_2^1)\end{aligned} \tag{33}$$

Accordingly, the specific optimility constraints of sub-optimization 2 are formulated as (34)–(36).

$$h_2(x,y_2) = 0,\ g_2(x,y_2) \le 0,\ y_2 \in [y^{\mathrm{L}}, y^{\mathrm{U}}], \tag{34}$$

$$\delta_2^1 \ge 0,\ \delta_2^1 \ge x - x_0,\ \nabla_{y_2} \mathscr{L}_4 = 0,\ {}^{1}\lambda_2, {}^{2}\lambda_2, {}^{3}\lambda_2 \ge 0, \tag{35}$$

$${}^{1}\lambda_2 \circ g_2(x,y_2) = 0,\ {}^{2}\lambda_2 \circ \delta_2^1 = 0,\ {}^{3}\lambda_2 \circ (x - x_0 - \delta_2^1) = 0 \tag{36}$$

where ${}^{1}\lambda_2$, ${}^{2}\lambda_2$, and ${}^{3}\lambda_2$ are the dual variables corresponding to inequality constraints.

Constrained sub-optimization 3: For the maximized optimization of the optimistic objective $\min_{y_2}[-F_2(x,y_2)]|_{x=x_0}$, the *Lagrange* function $\mathscr{L}_5$ is formulated as (37).

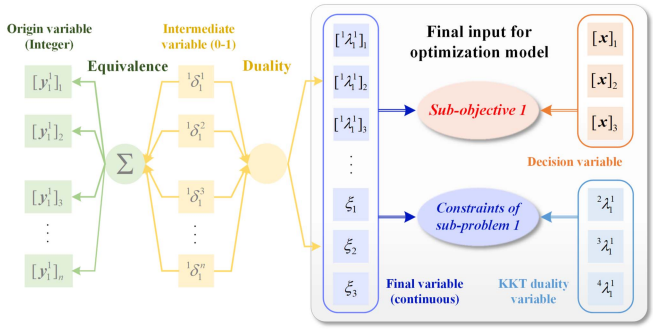


Fig. 3. Illustration of constrained sub-optimization 1.

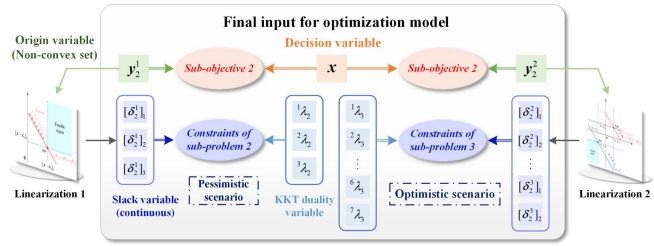


Fig. 4. Illustration of constrained sub-optimization 2 and 3.

$$\begin{aligned}\mathscr{L}_5 = &-|\delta_2^3|_1 + \upsilon_3{}^{\top} \cdot h_2(x,y_2) + {}^{3}\lambda_3{}^{\top} \cdot g_2(x,y_2) \\ &+ {}^{4}\lambda_3{}^{\top} \cdot (\delta_2^2 + x - x_0) + {}^{5}\lambda_3{}^{\top} \cdot (\delta_2^2 + x_0 - x) \\ &+ {}^{6}\lambda_3{}^{\top} \cdot (\delta_2^2 + \delta_2^3 - x + x_0) + {}^{7}\lambda_3{}^{\top} \cdot (\delta_2^2 + \delta_2^3)\end{aligned} \tag{37}$$

Accordingly, the specific optimility constraints of sub-optimization 3 are formulated as (38)–(43).

$$h_2(x,y_2) = 0,\ g_2(x,y_2) \le 0,\ y_2 \in [y^{\mathrm{L}}, y^{\mathrm{U}}] \tag{38}$$

$$\delta_2^2 \le x - x_0,\ \delta_2^2 \le x_0 - x,\ \delta_2^3 \le x - x_0 - \delta_2^2,\ \delta_2^3 \le -\delta_2^2 \tag{39}$$

$$\nabla_{y_2} \mathscr{L}_4 = 0,\ \ \nabla_{\delta_2^2} \mathscr{L}_5 = 0, {}^{1}\lambda_3, {}^{2}\lambda_3, {}^{3}\lambda_3, {}^{4}\lambda_3, {}^{5}\lambda_3 \ge 0 \tag{40}$$

$${}^{1}\lambda_3 \circ (\delta_2^2 + x - x_0) = 0, {}^{2}\lambda_3 \circ (\delta_2^2 + x_0 - x) = 0, \tag{41}$$

$${}^{3}\lambda_3 \circ g_2(x,y_2) = 0, {}^{4}\lambda_3 \circ (\delta_2^2 + x - x_0) = {}^{5}\lambda_3 \circ (\delta_2^2 + x_0 - x) = 0, \tag{42}$$

$${}^{6}\lambda_3 \circ (\delta_2^2 + \delta_2^3 - x + x_0) = 0, {}^{7}\lambda_3 \circ (\delta_2^2 + \delta_2^3) = 0, \tag{43}$$

where ${}^{3}\lambda_3$, ${}^{4}\lambda_3$, ${}^{5}\lambda_3$, ${}^{6}\lambda_3$, and ${}^{7}\lambda_3$ are the dual variables corresponding to inequality constraints.

Figs. 3 and 4 illustrate the linearization and constrained formulation processe for sub-optimizations 1, 2, and 3. (*i*) For the sub-optimization 1, the original optimization of variable y_1^1 within discrete uncertain set is firtstly reformulated to the equivalent binary optimization $(\delta_1^1, \delta_1^2, \ldots, \delta_1^{n-1}, \delta_1^n)$; Secondly, the dualization and constraint linearization of are developed to dualize the the binary optimization to be linear-convex optimization $({}^{1}\lambda_1, \xi)$; Finally, the *Lagrange* dual $({}^{2}\lambda_1, {}^{3}\lambda_1, {}^{4}\lambda_1)$ and KKT condition are introduced to establish the optimality conditions of sub-optimization 1. (*ii*) For example as sub-optimization 3, the

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TABLE I
CASE SETUP FOR SOLUTION PERFORMANCE COMPARISON

Case	Optimization approach	Wind speed distribution	Price distribution	Std	Interval
Case 1	Robust optimization	*Uniform*	*Uniform*	—	±10%
Case 2	Budget-set RO	*Uniform*	*Uniform*	—	±10%
Case 3	Stochastic optimization	*Gaussian*	*Uniform*	6.6%	±10%
Case 4	Scenario-based optimization	*Gaussian*	*Uniform*	6.6%	±10%
Case 5	NSIO	*Uniform*	*Uniform*	—	±10%

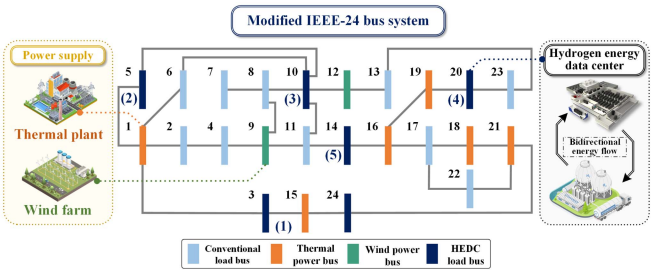


Fig. 5. Modified IEEE-24 bus system.

preferenced constraint are linearized by introducing intermediate variables δ_2^2 and δ_2^3. In this context, the value of variable δ_2^2 can be limited through KKT condition $({}^1\lambda_3, {}^2\lambda_3)$ to achieve the equivalence of linearization. The mathematical logic is visible in Linearzation 3 of Appendix A. On this basis, the *Lagrange* dual $({}^3\lambda_3, \ldots, {}^7\lambda_3)$ and KKT condition are introduced to establish the optimality conditions of sub-optimization 3.

Accordingly, the explicit mathematical model for NSIO is presented as (44).

$$\begin{aligned} \min_{x,y_1^1,y_2^1,y_1^2,y_2^2} \quad & [M(x,y_1^1,y_2^1,y_1^2,y_2^2),\ W(x,y_1^1,y_2^1,y_1^2,y_2^2)] \\ \text{s.t.} \quad & (1)-(21),\ (24)-(27),\ (33)-(A.2). \end{aligned} \tag{44}$$

Overall, the constrained uncertain sub-optimizations are the key for implementing the IO concept. The formulation of optimality constraints causes the variable quadraticity, i.e., mixed-integer linear programming. Due to the convex linearization for the original model, this complexity would not lead to unsolvability for this explicit mathematical IO model. In other words, the classical branch-and-bound algorithm is sufficient for optimizing the global optimal solution. For the bi-objective optimization, the augmented ϵ-constraint algorithm is applied to obtain the non-dominated solution set [18].

V. CASE STUDY

A. Case Study Setup

To validate the superiority of the NSIO approach and its solution performance, this section designs five cases and corresponding comparisons. Detailed explanations are provided in Subsection B. Considering the uncertain factors, Subsection C analyzes the economic effectiveness of the HEDC-UES strategy. Furthermore, the modified IEEE 24-bus system is used for the case study [35], and the specific topology diagram is visible in the Fig. 5. Due to space limitations, the specific parameters related to HEDCs, WP plants, and the power system are documented on: https://github.com/Xinxin-Long/NSIO_case.

TABLE II
CASE SETUP FOR ECONOMIC EFFICIENCE ANALYSIS

Case	Wind power fluctuation	Price fluctuation	Experimental	Control	DR program
Case 6	0%	0%	HEDC-UES	TUCD	Incentived
Case 7	3%	0%	HEDC-UES	TUCD	Incentived
Case 8	5%	5%	HEDC-UES	TUCD	Incentived

The case study setup for solution performance comparison is illustrated in Table I. Specifically, Cases 1 and 2 are respectively based on classical robust optimization (RO) and budget-set robust optimization [34]. Case 3 employs stochastic optimization (SO), where wind speed probability is modeled based on the *Gaussian* distribution. The standard deviation (STD) indicates that the uncertain fluctuation within this distribution is ±10%, consistent with other cases. Case 4 corresponds to scenario-based optimization. Here, solutions are generated based on established uncertain parameters, and the optimal solution is selected through performance comparison across multiple uncertain scenarios. Case 5 represents the proposed NSIO approach. Additionally, 1000 uncertain scenarios are sampled according to the Gaussian distribution. These scenarios are used for scenario-based optimization and as a test set for unified performance evaluation to validate the superiority of the proposed approach. The detailed model definition can be found in Appendix B.

To analyze the economic efficiency, this section also designs comparative groups to assess the effectiveness of decisions derived from the HEDC-UES strategy. The decision corresponding to power system pre-scheduling are set as the benchmark for comparison. This pre-scheduling is the thermal unit commitment dispatch (TUCD) without implementing the demand response program. The details of the case setup are provided in Table II.

B. Analysis of Solution Performance

The optimization results of NSIO, represented by the bi-objective *Pareto* solution set, are shown in Fig. 6. Under the uncertain optimization conditions, the optimal boundary for the

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TABLE III
NUMERICAL RESULTS OF SOLUTION PERFORMANCE COMPARISON

Indicator	Case 1	Case 2	Case 3	Case 4	Case 5
Mean	4.68	4.29	4.61	3.99	**3.90**
Median	4.73	4.34	4.64	4.04	**3.93**
STD	1.08	1.09	**1.02**	1.09	1.06
Width	3.65	3.73	3.49	3.66	**3.47**

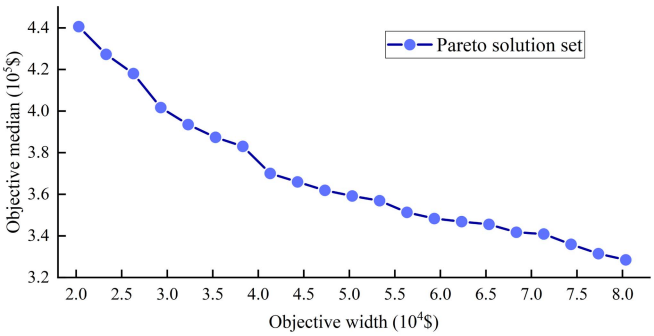


Fig. 6. bi-objective *Pareto* solution set of HEDC-UES.

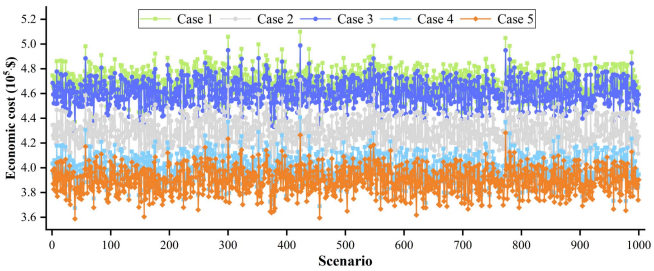


Fig. 7. Test result of solution performances.

cost-objective median is 3.29×10^5\$, and the optimal boundary for the cost-objective width is 2.03×10^4\$. For this minimization problem, the *Pareto* frontier exhibits a smooth convex shape, indicating that no solution completely dominates the others. In other words, if one solution performs better in terms of median cost, it will have worse performance in terms of cost width.

Based on the evaluation of 1000 random scenarios, the performance of the solutions for the five cases is presented in Fig. 7. For *Pareto* solution set, the solution corresponding to best performance in this evaluation is exhibited, similar to other cases. In this figure, the horizontal axis represents the sequence number of the random scenarios, while the vertical axis represents the economic cost. It is evident that the solution performance for Case 5, utilizing the NSIO approach, exhibits superior line distribution compared to the other cases. Specifically, the worst cost-objective bound of Case 5 is valued as 4.28×10^5\$ within Scenario 773. It has the less numerical value than Case 1: 5.10×10^5\$, Case 2: 4.71×10^5\$, Case 3: 4.99×10^5\$, and Case 4: 4.41×10^5\$, within the worst Scenario 423. On the other hand, the all best cost-objective bound is valued within Scenario 39. Specifically, this value of Case 5 is 3.59×10^5\$, which also has the less numerical value than Case 1: 4.36×10^5\$, Case 2: 3.96×10^5\$, Case 3: 4.29×10^5\$, and Case 4: 3.67×10^5\$. Therefore, the proposed NSIO approach provides solutions with a better lower-upper objective boundary compared to the other.

From the perspective of specific numerical results, Table III introduces the following four types of indicators to characterize the solution performance, i.e., Mean, Median, Standard deviation (STD), and Width. The NSIO solution corresponding to Case 5 achieves the optimal improvements compared with the Case 1, and the specific numerical values are listed as 20.0% of Mean, 20.4% of Median, 1.9% of STD, and 5.2% of Width. Being relative to Case 2, the improved performance of Case 5 can be presented as: 10.0% of Mean, 10.4% of Median, 2.8% of STD, and 7.5% of Width. The similar comparison with Case 3 reveals the Case 5 improvements as 18.2% of Mean, 18.1% of Median, and 0.6% of Width. For Case 4, the improvements of Case 5 solution are as 2.3% of Mean, 2.8% of Median, 2.8% of STD, and 5.5% of Width.

According to the analysis of the figures and numerical results, it can be concluded that the proposed NSIO demonstrates superior solution performance compared to classical RO, budget-set RO, and SO. For the mentioned scenario-based optimization, there are inherent advantages within its case setup. The reason is that the optimization-based scenario set is the same as the test set, which is an overly ideal condition. Nonetheless, the proposed NSIO still presents better solution performances. This result verifies the universality and optimality of the NSIO solution.

C. Analysis of Economic Efficiency

This section aims to analyze the economic efficiency of the proposed HEDC-UES strategy. Specifically, Table II presents three types of uncertainty scenarios, which are applied for evaluating the performance of two strategies: HEDC-UES and TUCD. The analysis includes HEDC economic costs and wind power utilization in the power system.

Fig. 8 illustrates the time-sharing electricity costs and wind power consumption. Case 6 is based on standard scenarios without fluctuations in WP and electricity prices. Compared to TUCD, the HEDC-UES decision achieves a cost reduction of 1.65×10^5\$ and an increase in WP utilization of 61.24×10^4 kW · h. In Case 7, the uncertainty fluctuation rate is set as 3%, corresponding to a 3% increase in wind speed and 0% increase in electricity price. Compared to TUCD, the HEDC-UES decision achieves a cost reduction of 1.65×10^5\$ and an increase in WP utilization of 240.36×10^4 kW · h. In Case 8, the uncertainty fluctuation rate is set as 5%, corresponding to a 5% increase in wind speed and a 5% increase in electricity price. Compared to TUCD, the HEDC-UES decision achieves a cost reduction

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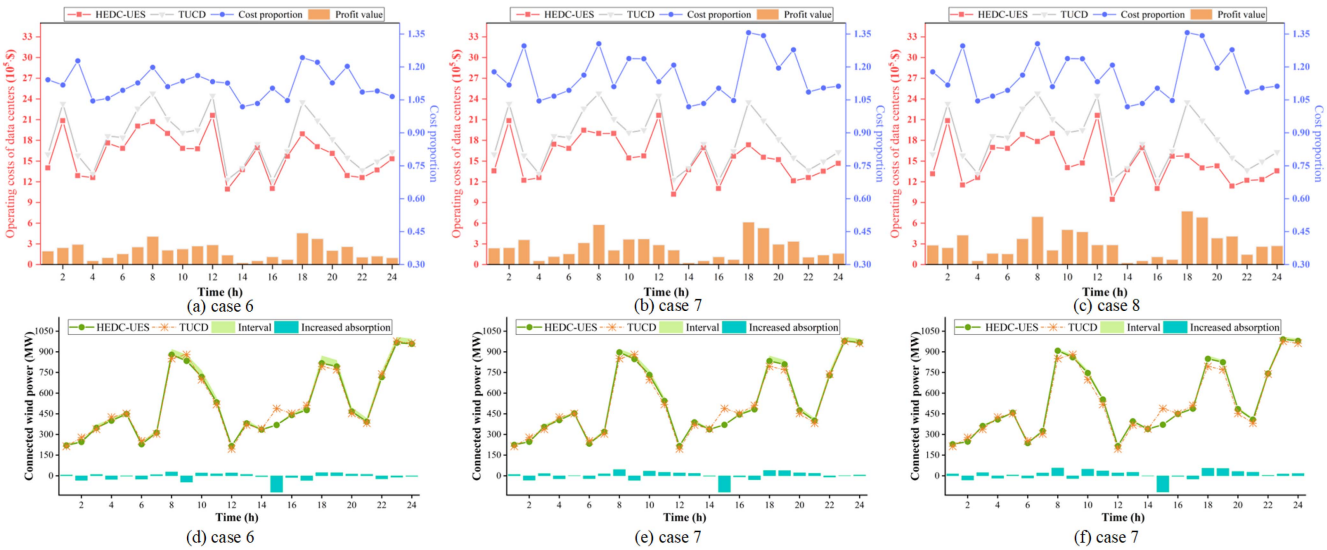


Fig. 8. Analysis of time-sharing electricity costs and wind power consumption.

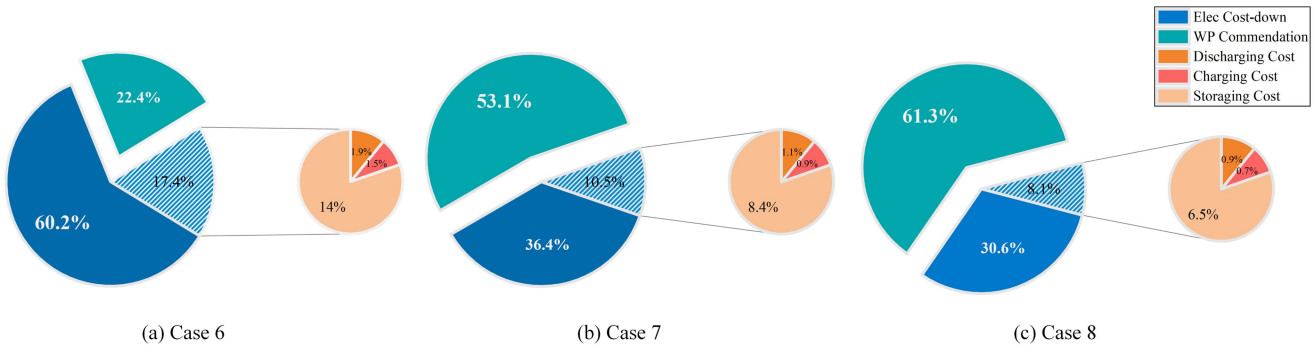


Fig. 9. Profit proportion for economic analysis.

of 1.80×10^5\$ and an increase in WP utilization of 360.91×10^4 kW · h. These comparison results indicate that: with the increase of available wind power, the HEDC-UES decision not only optimizes HEDC operational costs but also increases WP utilization in the power system. This demonstrates a win-win uncertainty optimization strategy for both HEDC operators and power system dispatchers.

From the perspective of HEDC economic optimization, Fig. 9 illustrates the proportion of optimized profits in three cases. These include electricity cost reduction, WP utilization incentives, and the cost increase of HES operation (negative profit). As the uncertainty fluctuation rate increases, HEDCs more actively respond to WP utilization, leading to higher utilization quantities and economic incentives. Consequently, the proportion of economic incentives increases from 22.4% to 61.3%, as shown in Fig. 9. On the other hand, despite the uncertain condition of electricity price increases, HEDCs still achieve a reduction in electricity costs. This is attributed to the flexible HEDC load shifting to avoid peak periods of electricity prices, as well as the superior adaptability of the HEDC-UES decision for uncertain scenarios. Therefore, these results sufficiently validate the effectiveness of the HEDC-UES strategy.

VI. Conclusion

This paper proposes an uncertain economic scheduling strategy to enhance the economic efficiency of hydrogen energy data centers and wind power utilization. From the perspective of the solving approach, the non-convex set interval optimization is explicitly formulated to obtain the optimal solution set under uncertain conditions. The specific conclusions are summarized as follows.

1) The uncertain economic scheduling strategy for hydrogen energy data centers effectively optimizes economic costs amidst the uncertainties of wind power supply and electricity prices. Additionally, it serves as a win-win strategy by enhancing flexibility and wind power utilization within the power system.
2) The non-convex set interval optimization demonstrates superior solution performance compared to well-recognized

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uncertain optimization approaches. This advantage can be attributed to the efficiency-preferred concept of interval optimization and the explicit mathematical modeling that ensures optimality.

The proposed uncertain optimization approach is broadly applicable to diverse power dispatch problems. In future work, related research will focus on models with coupling uncertain variables. It is a more challenging non-convexity. Decoupling strategies, explicit mathematical modeling, and feasible optimal solving method constitute our core concerns.

Appendix A
Proof of Specific Linearizations

Linearization 1: For the maximized or minimized optimization of binary objective $F_1(x, \delta_1^1, \delta_1^2, \ldots, \delta_1^{n-1}, \delta_1^n)|_{x=x_0}$, the vectors of dual variable ${}^1\lambda_1$ and intermediate variable ξ, $\xi = [\xi_1, \xi_2, \ldots, \xi_{\mathrm{N}\times n}]^\top$, are introduced. Accordingly, the linear-convex dual optimization is formulated as (A.1).

$$\begin{aligned} \max_{\lambda_1^1} \quad & F_1(x_0, {}^1\lambda_1, \xi) = -b^\top {}^1\lambda_1 + \sum_{\xi_i \in \xi} \xi_i \\ \text{s.t.} \quad & \xi_i \le \pm f_1^i + a_i^{\top 1}\lambda_1,\ \xi_i \le 0 \\ & {}^1\lambda_1 \ge 0, \quad i = 1, 2, \ldots, \mathrm{N} \times n. \end{aligned} \tag{A.1}$$

Proof: For the binary optimization (18), the *Lagrange* function $\mathscr{L}_1$ is formulated as (A.2).

$$\begin{aligned} \mathscr{L}_1(\delta_1, \lambda_1^1, \nu_1) &= {f_1}^\top \cdot \delta_1 \circ y_1^n + {}^1{\lambda_1}^\top \cdot (\mathrm{A} \cdot \delta_1 - b) + {\delta_1}^\top \cdot diag(\nu_1) \cdot \delta_1 \\ &= {\delta_1}^\top \cdot diag(\nu_1) \cdot \delta_1 + (f_1 + \mathrm{A}^\top \cdot {}^1\lambda_1 - \nu_1)^\top \cdot \delta_1 - b^\top \cdot {}^1\lambda_1, \end{aligned} \tag{A.2}$$

where ν_1 represents dual variable vectors corresponding to quadratic equality constraints. For this quadratic matrix function $\mathscr{L}_1(\delta_1, \lambda_1^1, \nu_1)$, the minimal function $g({}^1\lambda_1, \nu_1)$ is further transformed based on the division of the feasible region of variable δ_1, as presented in (A.3).

$$g({}^1\lambda_1, \nu_1) = \begin{cases} -b^\top \cdot {}^1\lambda_1 - \sum\limits_{i \in N\times n} \left(f_1^i + a_i^{\top 1}\lambda_1 - \nu_1^i\right)^2 / 4\nu_1^i, & \forall \nu_1^i \ge 0 \\ -\infty, & \exists \nu_1^i < 0 \end{cases} \tag{A.3}$$

Specifically, if $\forall \nu_1^i \ge 0$, this quadratic matrix function $\mathscr{L}_1(\delta_1, \lambda_1^1, \nu_1)$ is positive definite, thus the minimum function $g({}^1\lambda_1, \nu_1)$ would be valued at the first derivative point, $g({}^1\lambda_1, \nu_1) = -b^\top \cdot {}^1\lambda_1 - \sum_{i \in N\times n} \left(f_1^i + a_i^{\top 1}\lambda_1 - \nu_1^i\right)^2 / 4\nu_1^i$; If $\exists \nu_1^i < 0$, the quadratic matrix function is non-positive definite, thus the extreme value of minimum function is $g({}^1\lambda_1, \nu_1) = -\infty$. Accordingly, the dual formulation of the original optimization could be presented as (A.4).

$$\begin{aligned} \max_{\lambda_1^1, \nu_1^i} \quad & -b^\top \cdot {}^1\lambda_1 - \sum_{i \in N\times n} \left(f_1^i + a_i^{\top 1}\lambda_1 - \nu_1^i\right)^2 / 4\nu_1^i \\ \text{s.t.} \quad & \nu_1^i \ge 0 \end{aligned} \tag{A.4}$$

For variable ν_1^i, the upper bound of hyperbolic function as (A.4) is $\sup_{\nu_1^i \ge 0}[-(f_1^i + a_i^{\top 1}\lambda_1 - \nu_1^i)^2/4\nu_1^i] = (-|f_1^i + a_i^{\top 1}\lambda_1| + f_1^i + a_i^{\top 1}\lambda_1)/2$. On this basis, this explicit upper bound could be established as the piecewise function (A.5).

$$\begin{aligned} \sup_{\nu_1^i \ge 0}[-(f_1^i + a_i^{\top 1}\lambda_1 - \nu_1^i)^2/4\nu_1^i] &= \begin{cases} f_1^i + a_i^{\top 1}\lambda_1, & f_1^i + a_i^{\top 1}\lambda_1 \le 0 \\ 0, & f_1^i + a_i^{\top 1}\lambda_1 > 0 \end{cases} \\ &= \min\{0, f_1^i + a_i^{\top 1}\lambda_1\}, \end{aligned} \tag{A.5}$$

According to this piecewise function, if $f_1^i + a_i^{\top 1}\lambda_1 \le 0$, $\nu_1^i = -(f_1^i + a_i^{\top 1}\lambda_1)$ and $\delta_1^i = 1$; If $f_1^i + a_i^{\top 1}\lambda_1 > 0$, $\nu_1^i = f_1^i + a_i^{\top 1}\lambda_1$ and $\delta_1^i = 0$. In this context, the dual formulation of the original optimization could be reformulated as (A.6).

$$\begin{aligned} \max_{\lambda_1^1, \nu_1^i} \quad & -b^\top \cdot {}^1\lambda_1 - \sum_{i \in N\times n} \min\{0, f_1^i + a_i^{\top 1}\lambda_1\} \\ \text{s.t.} \quad & {}^1\lambda_1 \ge 0 \end{aligned} \tag{A.6}$$

It is worth noting that the optimimal value will be non-boundary, if ${}^1\lambda_1 < 0$. Thus, the constraint of ${}^1\lambda_1 \ge 0$ is supplemented. Furthermore, the preferenced constraint $\min\{0, f_1^i + a_i^\top \lambda_1^1\}$ is able to be equivalently linearized by introducing the vector of intermediate variable $\xi = [\xi_1, \xi_2, \ldots, \xi_{N\times n}]^\top$, as finally presented in (A.7).

$$\begin{aligned} \max_{\lambda_1^1} \quad & F({}^1\lambda_1, \xi) = -b^\top \cdot {}^1\lambda_1 + \sum_{\xi_i \in \xi} \xi_i \\ \text{s.t.} \quad & \xi_i \le f_1^i + a_i^{\top 1}\lambda_1,\ \xi_i \le 0 \\ & {}^1\lambda_1 \ge 0, \quad i = 1, 2, \ldots, N \times n. \end{aligned} \tag{A.7}$$

The minimum preference in this maximization optimization is similar to **Linearization 2**. Therefore, it would be explained in the follows. ■

Linearization 2: For the minimized optimization of the pessimistic objective $\max_{y_2}[-F_2(x, y_2)]|_{x=x_0}$, the vector of intermediate variable δ_2^1 is introduced. Accordingly, the linear-convex equivalent optimization is formulated as (A.8).

$$\begin{aligned} \min_{y_2} \quad & F_2(x_0, y_2) = |\delta_2^1|_1 \\ \text{s.t.} \quad & \delta_2^1 \ge 0, \delta_2^1 \ge x - x_0 \\ & h_2(x, y_2) = 0,\ g_2(x, y_2) \le 0, \\ & y_2 \in [y^{\mathrm{L}}, y^{\mathrm{U}}] \end{aligned} \tag{A.8}$$

Proof: For the non-convex objective involving preferenced constraint $F_2(x, y_2) = |\max\{0, x - x_0\}|_1$, the selection value of each preferenced item is defined as the vector of intermediate variable, i.e., $F_2(x_0, y_2) = |\delta_2^1|_1$. For the minimal first-order norm $|\delta_2^1|_1$, the constraint is supplemented as (A.9).

$$\delta_2^1 \ge 0,\ \delta_2^1 \ge x - x_0 \tag{A.9}$$

Specifically, the preferenced value within $\max\{0, x - x_0\}$ is the only valid lower bound constraint for δ_2^1, and the lower bound corresponding to non-preferenced value is failed. On the other hand, the minimum value of first-order norm $|\delta_2^1|_1$, optimal objective value, will be obtained at its unique extreme point, the above preferenced value. Therefore, the equivalence between (A.8) and the original optimization $\min_{y_2}[F_2(x, y_2)] = |\max\{0, x - x_0\}|_1$ is proved. ■

Linearization 3: For the maximized optimization of the optimistic objective $\min_{y_2}[-F_2(x, y_2)]|_{x=x_0}$, the vectors of intermediate variable δ_2^2 and δ_2^3 are introduced, and the *Lagrange*

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function $\mathscr{L}_2$ for variable δ_2^2 is defined as (A.10).

$$\mathscr{L}_2 = -|\delta_2^2|_1 + {}^1\lambda_3{}^\top \cdot (x - x_0) + {}^2\lambda_3{}^\top \cdot (x_0 - x). \quad \text{(A.10)}$$

Accordingly, the linear-convex equivalent optimization is formulated as (A.11).

$$\begin{aligned} \max_{y_2} \quad & F_2(x_0, y_2) = |\delta_2^3|_1 \\ \text{s.t.} \quad & \nabla_{\delta_2^2}\mathscr{L}_2 = 0, {}^1\lambda_3, {}^2\lambda_3 \geq 0, \\ & {}^1\lambda_3 \circ (\delta_2^2 + x - x_0) = 0, {}^2\lambda_3 \circ (\delta_2^2 + x_0 - x) = 0, \\ & \delta_2^2 \leq x - x_0, \delta_2^2 \leq x_0 - x, \delta_2^3 \leq x - x_0 - \delta_2^2, \delta_2^3 \leq -\delta_2^2 \\ & h_2(x, y_2) = 0,\ g_2(x, y_2) \leq 0,\ y_2 \in [y^{\mathrm{L}}, y^{\mathrm{U}}] \end{aligned} \quad \text{(A.11)}$$

Proof: For the non-convex objective involving preferenced constraint $F_2(x, y_2) = |\max\{0, x - x_0\}|_1$, the selection value of each preferenced item is defined as the vector of intermediate variable, i.e., $F_2(x_0, y_2) = |\delta_2^3|_1$. For the minimal first-order norm $|\delta_2^3|_1$, the constraint is supplemented as (A.12)–(A.13).

$$\sup[\delta_2^2] \leq x_0 - x,\ \sup[\delta_2^2] \leq x - x_0, \quad \text{(A.12)}$$

$$\delta_2^3 \leq x - x_0 - \delta_2^2,\ \delta_2^3 \leq -\delta_2^2, \quad \text{(A.13)}$$

Specifically, the preferenced value of $\min\{x - x_0, x_0 - x\}$ is the only valid upper bound constraint for δ_2^2, $\sup[\delta_2^2] = \min\{x - x_0, x_0 - x\} \leq 0$. If sub-optimization $\sup[\delta_2^2]$ is established, the supplemented constraint (A.13) is reformulated as (A.14).

$$\delta_2^3 \leq x - x_0 + \max\{x - x_0, x_0 - x\}, \delta_2^3 \leq \max\{x - x_0, x_0 - x\}, \quad \text{(A.14)}$$

Accordingly, if $x - x_0 \geq 0, \max\{x - x_0, x_0 - x\} = x - x_0$ is the only valid upper bound constraint for δ_2^3; if $x - x_0 \leq 0$, $x - x_0 + \max\{x - x_0, x_0 - x\} = 0$ is the only valid upper bound constraint for δ_2^3. It is equivalent to $\delta_2^3 \leq \max\{0, x - x_0\}$. On the other hand, the modeling of $\sup[\delta_2^2]$ could be formulated as optimility constraints as (A.15)–(A.16).

$$\nabla_{\delta_2^2}\mathscr{L}_2 = 0, {}^1\lambda_3, {}^2\lambda_3 \geq 0, \quad \text{(A.15)}$$

$${}^1\lambda_3 \circ (\delta_2^2 + x - x_0) = 0, {}^2\lambda_3 \circ (\delta_2^2 + x_0 - x) = 0, \quad \text{(A.16)}$$

On this basis, the maximum value of first-order norm $|\delta_2^3|_1$, optimal objective value, will be obtained at this unique extreme point, the above preferenced value. Therefore, the equivalence between (A.11) and the original optimization $\max_{y_2}[F_2(x, y_2)] = |\max\{0, x - x_0\}|_1$ is proved. ■

Appendix B
Model Definations of the Benchmarks in Table I

Robust optimization of Case 1: In the experiments, *Lagrangian* duality is used to formulate the robust optimization model. The following presents the key model derivations for the formulations of the robust optimization. Firstly, for integer uncertain sets, the y_1 associated constraints are presented as (B.1).

$$y_1 : \alpha_t^1 + (\alpha_t^{n,\max} - \alpha_t^1) \cdot L_{i,t}^{\mathrm{Rate}} - \alpha_t \leq 0, \quad \text{(B.1)}$$

To ensure global constraint feasibility, the objective of the robust sub-optimization should be formulated as $\sup[-\alpha_t]$. Furthermore, the integer variables α_t can be transformed as the binary by (17). Thus, the correlation coefficients is characterized by $\mathrm{A}_t'(x)$ and b_t', which involve the transformation constraints (17) and the normal constraints in (18). Notably, to standardize the model, all subsequent constrained optimization models are formulated in matrix form. Accordingly, the robust sub-optimization can be specifically described as (B.2).

$$\begin{aligned} \min \quad & y_{1,t}^\top \cdot \delta_{1,t} \\ \text{s.t.} \quad & \mathrm{A}_\mathrm{t}'(\mathrm{x}) \leq \mathrm{b}_\mathrm{t}'(\mathrm{x}, \delta_1^1,\ \delta_1^2,\ \ldots,\ \delta_1^{\mathrm{n}-1},\ \delta_1^{\mathrm{n}}), \end{aligned} \quad \text{(B.2)}$$

where $[\cdot]_n$ denotes the n_{th} element in the column vector., $y_{1,t} = \{[y_1^t]_1, [y_1^t]_2, \ldots, [y_1^t]_n\}^\top$, and $\delta_{1,t} = \{[\delta_1^t]_1, [\delta_1^t]_2, \ldots, [\delta_1^t]_n\}^\top$. According to **Linearization 1**, the equivalent dual formulation of the binary model can be expressed as:

$$\begin{aligned} \max_{{}^1\lambda_1'} \quad & f_t^{\mathrm{RO}}(x_1, {}^1\lambda_{1,t}', \xi_t') = -b_t'{}^\top \cdot {}^1\lambda_{1,t}' + \sum_{\xi_{i,t}' \in \xi_t'} \xi_{i,t}' \\ \text{s.t.} \quad & \xi_{i,t}' \leq [y_1^t]_i + a_{i,t}'{}^\top \cdot {}^1\lambda_{1,t}',\ \xi_{i,t}' \leq 0 \\ & {}^1\lambda_{1,t}' \geq 0,\ \ i = 1, 2, \ldots, n. \end{aligned} \quad \text{(B.3)}$$

Similar to **Constrained sub-optimization 1**, the *Lagrangian* function for the single variable α_t, $\alpha_t = y_{1,t}^\top \cdot \delta_{1,t}$, can be expressed as:

$$\begin{aligned} \mathscr{L}_{3,t}' = & -b_t'{}^\top \cdot {}^1\lambda_{1,t}' + \sum_{\xi_{i,t}' \in \xi_t'} \xi_{i,t}' + \sum_{i \in n} {}^2\lambda_{1,t}^i{}^\top \cdot (\xi_{i,t}' \\ & - [y_1^t]_i - a_{i,t}'{}^\top \cdot {}^1\lambda_{1,t}') + {}^3\lambda_{1,t}{}^\top \cdot \xi_t' - \lambda_{1,t}'{}^\top \cdot {}^4\lambda_{1,t} \end{aligned} \quad \text{(B.4)}$$

where the vectors of dual variable ${}^1\lambda_1'$ and intermediate variable $\xi_t', \xi = [\xi_{1,t}', \xi_{2,t}', \ldots, \xi_{n,t}']^\top$ are introduced. The optimality constraints for integer robust sub-optimization can be formulated as (B.5)–(B.7).

$$\nabla_{\lambda_{1,t}^i, \xi_t'}\mathscr{L}_{3,t}' = 0,\ \xi_{i,t}' \leq [y_1^t]_i + a_{i,t}'{}^\top \cdot {}^1\lambda_{1,t}',\ \xi_{i,t}' \leq 0, \quad \text{(B.5)}$$

$${}^1\lambda_{1,t}, {}^2\lambda_{1,t}, {}^3\lambda_{1,t}, {}^4\lambda_{1,t} \geq 0,\ {}^1\lambda_{1,t}{}^\circ {}^4\lambda_{1,t} = 0,\ {}^3\lambda_{1,t}{}^\circ \xi_t' = 0, \quad \text{(B.6)}$$

$${}^2\lambda_{1,t}^i{}^\circ([y_1^t]_i + a_{i,t}'{}^\top \cdot {}^1\lambda_{1,t}') = 0 \quad \text{(B.7)}$$

where ${}^2\lambda_1'$, ${}^3\lambda_1'$, and ${}^4\lambda_1'$ are the dual variables corresponding to inequality constraints.

For prefered uncertain sets, the y_2 associated constraints are presented as (B.8).

$$P_{w,t}^{\mathrm{WP}} - P_{w,t}^{\mathrm{AWP}} \leq 0,\ P_{w,t}^{\mathrm{AWP}} \in [P_{w,t}^{\mathrm{AWP,L}}, P_{w,t}^{\mathrm{AWP,U}}] \quad \text{(B.8)}$$

To ensure global constraint feasibility, the objective of the robust sub-optimization should be formulated as $\sup[-P_{w,t}^{\mathrm{AWP}}]$, and the optimization direction of which ensures that: the transformation of prefered sets adheres to **Linearization 2**. Therefore, the equivalent linearized model of the prefered set can be formulated as (B.9).

$$\begin{aligned} \min_{[y_2]_t} \quad & f_t^{\mathrm{RO}}(x_0, [y_2]_t) = [y_2]_t \\ \text{s.t.} \quad & [\delta_2^1]_t \geq 0,\ \ [\delta_2^1]_t \geq [x - x_0]_t \\ & h_2(x, [y_2]_t) = 0,\ g_2(x, [y_2]_t) \leq 0,\ y_2 \in [y^{\mathrm{L}}, y^{\mathrm{U}}] \end{aligned} \quad \text{(B.9)}$$

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Similar to **Constrained sub-optimization 2**, the *Lagrangian* function for the single variable $[y_2]_t$ can be expressed as:

$$\mathscr{L}_{4,t} = [y_2]_t + {\upsilon'_{2,t}}^{\top} \cdot h_2(x,[y_2]_t) + {}^1\lambda'_{2,t}{}^{\top} \cdot g_2(x,[y_2]_t) - {}^2\lambda'_{2,t}{}^{\top} \cdot [\delta_2^1]_t + {}^3\lambda'_{2,t}{}^{\top} \cdot ([x-x_0]_t - [\delta_2^1]_t) \tag{B.10}$$

where ${}^1\lambda'_{2,t}$, ${}^2\lambda'_{2,t}$, and ${}^3\lambda'_{2,t}$ are the dual variables corresponding to inequality constraints; $\upsilon'_{2,t}$ is the dual variable corresponding to equality constraints. Accordingly, the optimality constraints for integer robust sub-optimization can be formulated as (B.11)–(B.14).

$$h_2(x,[y_2]_t) = 0,\ g_2(x,[y_2]_t) \le 0,\ y_2 \in [y^{\mathrm{L}}, y^{\mathrm{U}}] \tag{B.11}$$

$$[\delta_2^1]_t \ge 0,\ \ [\delta_2^1]_t \ge [x-x_0]_t,\ \nabla_{[y_2]_t,[\delta_2^1]_t}\mathscr{L}_{4,t} = 0, \tag{B.12}$$

$${}^1\lambda'_{2,t} \cdot g_2(x,[y_2]_t) = 0,\ {}^2\lambda'_{2,t} \cdot [\delta_2^1]_t = 0, \tag{B.13}$$

$${}^3\lambda'_{2,t} \cdot ([x-x_0]_t - [\delta_2^1]_t) = 0,\ \ {}^1\lambda'_{2,t}, {}^2\lambda'_{2,t}, {}^3\lambda'_{2,t} \ge 0. \tag{B.14}$$

■

Budget-set robust optimization of Case 2: The main defference between the budget-set RO and the classical RO is that: the uncertain intervals of budget-set RO is limited by the established budget parameter Γ. This additional value constraint is to improve conservatism. In addition to the added parameter Γ and related budget-set constraints, the modeling strategy of budget-set RO is consistent with the classical one. To avoid redundant model repetitions, only critical constraint additions and model modifications are presented as follows.

i) Budget-set constraints for integer set

The specific budget constraints for integer variables can be expressed as (B.15).

$$\sum_{k=1}^{\mathrm{N}} (y_{1,k}^{\top} \cdot \delta_{1,k}) / [y_1^k]_n \ge \Gamma_1, \tag{B.15}$$

where $(y_{1,k}^{\top} \cdot \delta_{1,k})/[y_1^k]_n$ denotes the budget rate for single uncertain integer variable, and Γ_1 represents the total budget parameter. Thus all uncertain variables are additionally subjected to a new boundary restrictions. This addition means the extreme conservative scenarios can be avoided by adjusting the parameter Γ_1. Furthermore, the separation of variable and constants in this formula can be visible as (B.16)–(B.17).

$$(y_{1,t}^{\top} \cdot \delta_{1,t})/[y_1^t]_n \ge \Psi(\delta_1^1,\ \delta_1^2,\ \ldots,\ \delta_1^{n-1},\ \delta_1^n), \tag{B.16}$$

$$\Psi = \Gamma_1 - \sum_{k=1}^{t-1} (y_{1,k}^{\top} \cdot \delta_{1,k})/[y_1^k]_n - \sum_{k=t+1}^{\mathrm{N}} (y_{1,k}^{\top} \cdot \delta_{1,k})/[y_1^k]_n, \tag{B.17}$$

It is noteworthy that the robust modeling is applied to a single uncertain integer variable. Consequently, all integer variables except $\delta_{1,t}$ are treated as constants. Accordingly, the matrix parameters in Model (B.2) can be reformulated as (B.18).

$$\mathrm{A}''_t(x) = \begin{pmatrix} \mathrm{A}'_t(\mathrm{x}) \\ y_{1,t}^{\top}/[y_1^k]_n \end{pmatrix}, b''_t = \begin{pmatrix} b'_t(x,\delta_1^1,\delta_1^2,\ldots,\delta_1^{n-1},\delta_1^n) \\ \Psi(\delta_1^1,\delta_1^2,\ldots,\delta_1^{n-1},\delta_1^n) \end{pmatrix} \tag{B.18}$$

ii) Budget-set constraints for prefered set

The specific budget constraints for the variables involved prefered set can be expressed as (B.19).

$$\sum_{k=1}^{2\mathrm{N}} [y_2]_k / [y^{\mathrm{U}}]_n \ge \Gamma_2, \tag{B.19}$$

where $[y_2]_k/[y^{\mathrm{U}}]_n$ denotes the budget rate for single uncertain variable, and Γ_2 represents the total budget parameter. In the experiment, $\Gamma_1 = \mathrm{N}/2,\ \Gamma_2 = \mathrm{N}$.

Stochastic optimization of Case 3: In stochastic optimization, probability distributions and random sampling techniques are employed to characterize uncertainty. Consequently, during model optimization, the uncertain factors are set as known constants through external means above mentioned.

First, the table presents the assumed distributions for two categories of uncertain factors. Secondly, **Latin Hypercube Sampling** (LHS) is applied to generate random scenarios Y, $\mathrm{Y} = \{\mathrm{Y}^1, \mathrm{Y}^2, \ldots, \mathrm{Y}^{\mathrm{R}}\}$. Third, scenario reduction techniques are employed to select representative scenarios, with scenario dimensions of $\mathrm{R} \to \mathrm{r}$. Accordingly, the conceptual model for the stochastic optimization optimal solution $\bar{x}$ is formulated as (B.20).

$$\bar{x} = \arg\min_x \tfrac{1}{\mathrm{r}} \sum_{i=1}^{\mathrm{r}} F(x, \mathrm{Y}^i), \tag{B.20}$$

$\bar{x}$ corresponds to the optimal objective expectation for the all random scenarios Y.

Scenario-based optimization of Case 4: Classical stochastic optimization employs scenario reduction techniques to decrease the number of random scenarios involved in the optimization. This is implemented to prevent dimensional explosion of variables. In benchmark testing, the scenario-based optimization adopts a stepwise strategy to prevent dimensional explosion in any single model, while utilizing all sampled scenarios.

Specifically, the conceptual model for the solution set $\bar{x}^i$ of scenario-based optimization is formulated as (B.21).

$$\bar{x}^i = \arg\min_{x^i} F(x^i, \mathrm{Y}^i),\ \ i \in \{1,2,\ldots,\mathrm{R}\}. \tag{B.21}$$

The solutions $\bar{x}^i$ in the solution set are optimized based on the single scenario Y^{i}. Accordingly, the optimal solution $\bar{x}'$ is selected within this solution set, based on the mean comparison of objective value.

$$\bar{x}' = \arg\min_{\bar{x}^i} \tfrac{1}{\mathrm{R}} \left[\sum_{j=1}^{\mathrm{R}} F(\bar{x}^1, \mathrm{Y}^j), \sum_{j=1}^{\mathrm{R}} F(\bar{x}^2, \mathrm{Y}^j), \ldots, \sum_{j=1}^{\mathrm{R}} F(\bar{x}^i, \mathrm{Y}^j) \right] \quad i,j \in \{1,2,\ldots,\mathrm{R}\}. \tag{B.22}$$

For this stepwise optimization strategy, global optimality of solutions may not be guaranteed. However, its advantage is the comprehensive utilization of all sampled scenarios.

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